

KRISHNA KUMAR M

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EXPERIENCE

HCLTech | Graduate Engineer Trainee

Oct 2024 – Present | Chennai, Tamil Nadu

Cybersecurity — Email Security

- Triaged and analysed 500+ suspicious emails per week using **Cisco IronPort** and **Abnormal AI**, reducing phishing false-negative rate by 30% through custom policy tuning and rule configuration.
- Engineered spam-filtering rules and **secure mail-flow policies** across 3 enterprise client environments, cutting malicious email delivery incidents by 45% within 2 months of deployment.
- Led cross-functional threat-response drills with security and IT teams for 2 client accounts, achieving full **PCI-DSS email compliance** sign-off 3 weeks ahead of schedule.

Cloud & DevOps — GCP / AWS

- Designed and deployed cloud infrastructure on **GCP** (Compute Engine, IAM, Pub/Sub, Cloud Storage, Cloud SQL) for 3 client environments, reducing manual provisioning time by 40% through Infrastructure as Code (IaC).
- Implemented **IAM policies and VPC networking** controls on GCP to enforce least-privilege access across 5 service accounts, eliminating 100% of over-privileged roles flagged in a security audit.
- Automated cloud resource monitoring using **GCP Pub/Sub + Cloud Functions**, cutting average incident detection time from 25 minutes to under 5 minutes.

Tech stack: GCP (Compute Engine, IAM, Pub/Sub, Cloud SQL, Cloud Storage), AWS (EC2, S3, IAM), Cisco IronPort, Abnormal AI, Terraform

PROJECTS

AI-Driven Chatbot for Mental Health

Jan 2024 – Apr 2024

- Built a mental health support chatbot using **Python (Flask)** backend and **React.js** frontend, integrating the OpenAI GPT API to deliver context-aware, empathetic responses across 10+ mental health topics.
- Implemented end-to-end **AES-256 conversation encryption** and anonymised user session storage in **MySQL** to ensure full data privacy, achieving zero data-breach incidents across 30+ pilot users.
- Deployed on **GCP Cloud Run** with auto-scaling configured, achieving <200ms average response latency under concurrent load of 50+ users.
- Achieved **87% user satisfaction rating** in a structured pilot study (n=30) based on a post-session feedback survey, with 92% of users rating the chatbot "easy to use."

Tech stack: Python, Flask, React.js, OpenAI GPT API, MySQL, GCP Cloud Run, AES-256 encryption

Full-Stack Employee Task Management System

Aug 2023 – Dec 2023

- Designed and developed a full-stack **Spring Boot + React.js** task management application with role-based access control (RBAC) for 3 user roles (Admin, Manager, Employee), supporting 50+ concurrent users.
- Built 15+ **RESTful APIs** using Spring Boot and Hibernate ORM, reducing average API response time to under 80ms through query optimisation and connection pooling with **MySQL**.
- Implemented **JWT-based authentication** and session management, eliminating unauthorised access attempts across all user roles in testing.
- Containerised the application with **Docker**, enabling one-command local deployment and reducing environment setup time by 70%.

Tech stack: Java, Spring Boot, Hibernate, React.js, MySQL, REST APIs, JWT, Docker, Git

GCP Cloud Infrastructure Automation (IaC)

Sep 2023 – Nov 2023

- Automated provisioning of a multi-service GCP environment (Compute Engine VMs, VPC, Firewall rules, Cloud Storage buckets) using **Terraform**, reducing manual setup time from 4 hours to under 10 minutes.

- Created reusable **Terraform modules** for 4 environment types (dev, staging, prod, DR), ensuring 100% configuration parity and eliminating environment-drift issues.
- Integrated GCP **Cloud Monitoring + alerting policies** to auto-notify on CPU >80% and disk >75%, cutting unplanned downtime by an estimated 60%.

Tech stack: Terraform, GCP (Compute Engine, VPC, Cloud Storage, IAM, Cloud Monitoring), Bash scripting, Git

EDUCATION

KLN College of Engineering, Sivagangai

2020 – 2024

B.Tech — Information Technology | CGPA: **9.07 / 10**

Christ The King MHSS, Madurai

2020 (HSC)

Percentage: 75%

Christ The King MHSS, Madurai

2018 (SSLC)

Percentage: 90%

CERTIFICATIONS

Associate Cloud Engineer | Google Cloud Platform (GCP)

Feb 2024

TECHNICAL SKILLS

Languages: Java (Advanced), Python (Intermediate), C (Intermediate)

Frontend: React.js, HTML, CSS, JavaScript

Backend & Frameworks: Spring Boot, Hibernate, Node.js, Flask, REST APIs, Microservices

Databases: MySQL, PostgreSQL

Cloud & DevOps: GCP (Compute Engine, IAM, Pub/Sub, Cloud Storage, Cloud SQL, Cloud Run), AWS (EC2, S3, IAM), Terraform, Docker, Git

Security: Cisco IronPort, Abnormal AI, Email Threat Analysis, IAM Policy Management, AES-256 encryption

Testing: JUnit, Postman, REST API testing